



ELE101-1

Introduction

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Outline

Getting started

Survey

About Me

Curriculum

Schedule

Motivations

Tech. Part



Getting Started Survey

- No personal information is collected
- Think before answering
- Help us prepare better course content



About me

BSc, METU, 1998



MSc, AYBU, 2015



PhD, TOBB ETU, ...



TOBB ETÜ
Ekonomi ve Teknoloji Üniversitesi



Certificates

Certified Instructor and University
Ambassador at NVIDIA



GitHub Teacher



Work Experience

- ASELSAN
 - Türkiye's leading defence company
 - Aselsan 47. Rank in "Defense News' Top 100 list"
 - ~15 years
 - March '23



aselsan

Türk Silahlı Kuvvetlerini Güçlendirme Vakfı'nın bili. Kuruluşudur.



About Me

- 25 years in software development
- 15+ years in telecom field
- PhD thesis student @ TOBB ETÜ
- Lecturer @ TOBB ETÜ
- ELE361L (Telecom Lab)
 - 2021-2022 Summer
 - ...
 - 2025-2026 Fall ELE361L - marks the 7th time
- ELE468 Software Defined Radio (SDR)
- ELE476 Applied Digital Signal Processing
- ELE495 Senior Project

Embedded Experience - Monitoring Receivers

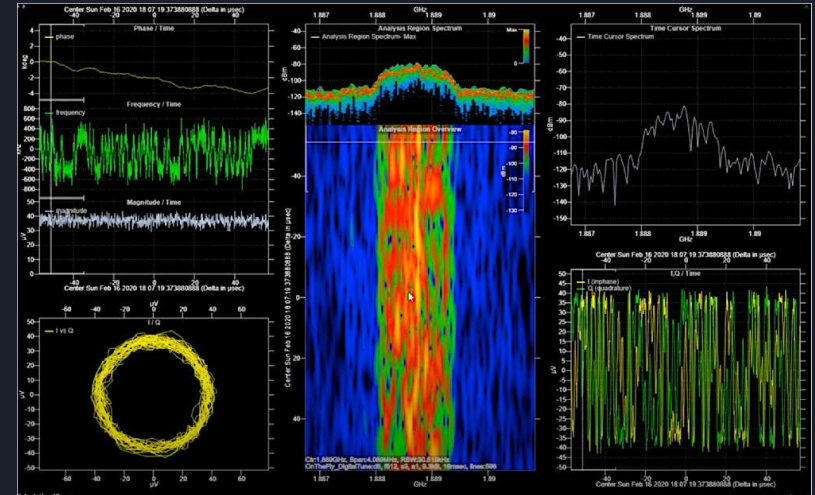
TI 8-core DSP/SysBIOS



Intel i7/VxWorks



- Offline/Online Analysis
- Demodulation/Decoding
- Parameters
 - Center Freq
 - Modulation Type
 - Baud Rate



ELE361L

ELE361L

Communication Systems Laboratory

Course Info - 5min 



Open System

Course was designed from the ground up to have open system components.



Modular

Course has modular architecture. It can be customized or extended by adding new modules.



Real-World Signals

We are dealing with real-world signals like AM in airband, broadcast FM signals, or ADS-B signals emitted from aircrafts.



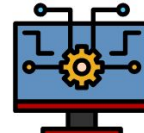
Professional Development

We support professional development of students in software. They learn industry-standard tools and improve their programming skills.



Mobility/Remote Opportunity

Course provides students mobility. They can complete their tasks wherever they feel comfortable. And it is also remote-ready.



Software-based

Since we are using Software Defined Radios, all modules are software-based.

<https://ele361l.github.io/>

Awards: 9. Başakşehir Innovation Contest



Awards: 3. AOSB R&D and Innovation Contest



3. Ulusal Sanayi Odaklı AR-GE ve İnovasyon Proje Yarışması

ÖDÜL KAZANANLAR
Akademisyen/Lisansüstü Mezun

BİRİNCİ
Mücahid KUTLU
Tematik Alan: Bilgisayar
100.000 TL

İKİNCİ
Mutlu KURBAN
Tematik Alan: Tekstil
80.000 TL

ÜÇÜNCÜ
Murat SEVER
Tematik Alan: Elektrik/Elektronik
60.000 TL

500.000 TL
Ödül Havuzu

The poster features a central illustration of a large golden trophy with the AOSB logo on it, surrounded by several smaller golden trophies. The background is dark with a pattern of small white stars and gears.

Events: SDR Academy Friedrichshafen, Germany



Events: GNU Radio Conference 2023



5-9 September 2023

Talk & Workshop



Events: GNU Radio Conference 2025

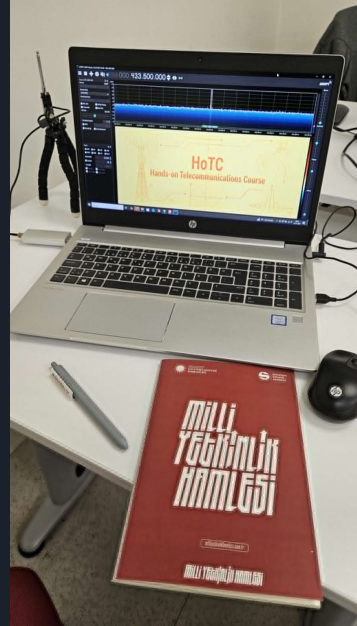


8-12 September 2025

Workshop: Simple Replay Attack Demo with
GNU Radio



SSA: RF School Coordinator



Reachout Project



Linux Camps

- Winter
 - 2025 Eskişehir
 - 2026 Afyonkarahisar
- Summer
 - Bolu
- All FREE



- Ağ Yönetimine Giriş
- Blokzincir'e Giriş ve Ethereum ile Akıllı Sözleşmeler
- C ve Assembly Programlama ile Linux Çekirdeğine Bir Bakış
- GNU/Linux Sistem Yönetimi 1. Düzey
- GNU/Linux Sistem Yönetimi 2. Düzey
- Linux İç Yapısı: (e)BPF'e Giriş
- PHP & Laravel & Vue.js ile Yapay Zeka Destekli Yazılım Geliştirme
- Python ile Programlamaya Giriş
- Python ile Yapay Zeka
- Sayısal Kriptografi ve TersKod Mühendisliğine Giriş

My Research Items

- <https://ele361l.github.io/blog>





More on LinkedIn

- I am on LinkedIn
- Get connected to hear more from me!





Office/Make-up Hours

- Office
 - Tuesday 15:00-16:30
 - Thursday 12:30-14:30
- Make-up
 - Saturday or Sunday?
 - Morning or Afternoon?



Content

1. Giriş ve Bölüm Hocalarının ve Program Sürecinin Tanıtımı (4 Saat – 2 hafta)
2. Mühendislik Tasarımı ve Mühendislik Etiği (4 Saat – 2 hafta)
3. Yazılı ve Sözlü Sunum Teknikleri (4 saat – 2 hafta)
4. Girişimcilik (2 saat – 1 hafta)
5. Python Dersleri (6 saat – 3 hafta)
6. Elektronik Mühendisliği Alt Alanlarından Sunumlar (4 saat – 2 hafta)



Assessment



Attendance 9/10 miss anything!
Assign you miss anything Reasoned



Key Points to Remember

Some important points about the course

- Always bring your laptop to class
- Every user must have an account on [GitHub](#)
- Make installments required for the course into your laptop ASAP
- Check your TOBB ETU e-mail for any updates!
- Do not forget to sign in the attendance list!
- Common time for compensation?



Success Strategies

- Make time
- Assignments and Deadlines
- Understand the technology
- Do Your Work yourself
- Save Your Work!
- Start Early

Motivations

- The Missing Semester of Your “EE” Education
- Engagement
- Professional Development





The Missing Semester of Your “EE” Education

- Advanced EE Topics
 - Circuit Theory
 - Electronics
 - Electromagnetics
 - Control Systems
 - Communication Systems
- But, wait! What about the tools?
 - Version Control System - Git
 - Command Shell or terminal - Cmd or Bash
 - Python
 - Remote Machines - serial, ssh
 - Virtual machines or containers

The Missing Semester of Your CS Education



Schedule

1. GitHub
2. Markdown
3. CLI: Command Prompt/Bash Prompt
4. Linux
5. git
6. conda
7. JupyterLab
8. Python
 - a. Basics
 - b. numpy, matplotlib
 - c. hexadecimal, dB Calculation, complex numbers
9. ssh
10. docker



Skills

- Technical Skills
- Soft Skills



Technical Skills

- **Programming and Software Proficiency:**
 - Familiarity with programming languages (e.g., Python, C/C++, MATLAB) and industry-specific software (e.g., AutoCAD, SolidWorks, ANSYS).
- **Mathematics and Data Analysis:**
 - Proficiency in calculus, linear algebra, and statistical analysis to model systems and interpret data.
- **Design and CAD (Computer-Aided Design):**
 - Skills in creating and modifying 2D and 3D models for prototypes, simulations, and manufacturing processes.
- **Project Management and Planning:**
 - Competence in using tools like Microsoft Project or Primavera for managing timelines, resources, and budgets in engineering projects.
- **Electrical System Knowledge:**
 - Understanding of electrical components, circuits, and systems for interdisciplinary projects.
- **Testing and Quality Assurance:**
 - Ability to design, execute, and interpret tests to ensure that products meet safety, functionality, and reliability standards.



Soft Skills

- Soft Skills
 - Critical Thinking
 - Teamwork
 - Leadership
 - Being Positive
 - Problem Solving
 - Communication
 - Awareness

Savunma Sanayi Akademisi

7 Temel Sosyal Yetkinlik

<https://ssa.gov.tr/katalog/?id=5#book1/2>



ELEŞTİREL DÜŞÜNME

(Akıl yürütme, analiz ve değerlendirme gibi zihinsel süreçlerin tamamını kapsar.)



TAKIM ÇALIŞMASI

(Ekip ruhunu anlayabilme, uyumlu çalışabilme, birbirini yönetebilme, başkalarının çalışmalarını planlamayı ve organize edebilmeyi kapsar.)



LİDERLİK

(Liderlik, sadece istenileni yaptırmak değil insanlara dokunabilmek ve ilham olabilmektir; özgüven ve vizyon sahibi olabilmeyi kapsar.)



POZİTİF YAKLAŞIM & DAYANIKLILIK

(İniş çıkışlarla dolu iş dünyasında karamsarlık girdabına kapılmamak büyük önem taşımaktadır. Pozitif yaklaşım, sorunları çözmeyi, engelleri aşmayı ve başarıya giden yeni yollar yaratmayı kapsar.)



PROBLEM ÇÖZME & VERİ YÖNETİMİ

(Yaratıcı olmayı, inovatif düşünmeyi, öğrenmeye ve araştırmaya açık olmayı ve pes etmeme becerilerinin bütünü kapsar.)



KÜLTÜREL FARKINDALIK & İFADE YETKİNLİĞİ

(Çeşitli kitle iletişim araçları kullanılarak görüş, deneyim ve duyguların yaratıcı bir şekilde ifade edilmesine yönelik becerileri kapsar.)



SİBER FARKINDALIK

(Siber farkındalık, bireylerin ve kurumların çevrimiçi olarak karşılaşılabilecekleri riskleri anlamalarını ve bunlara karşı korunmalarını sağlar. Siber tehditleri tanıma, önleme ve müdahale etme yeteneklerinin tümünü kapsar.)

Kaynak: EDSP - Vision on defence related skills for Europe today and tomorrow / 2019



How to use GitHub?

Web interface

<https://github.com/>

Create an account if not yet!

Do not forget your account info!



Markdown Format

- <https://docs.github.com/en/get-started/writing-on-github/getting-started-with-writing-and-formatting-on-github/basic-writing-and-formatting-syntax>



Create your first repo!

- Add README
- Add a photo
- Download as zip
- Clone



Thanks!
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